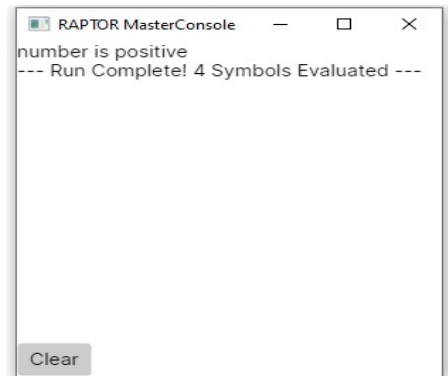
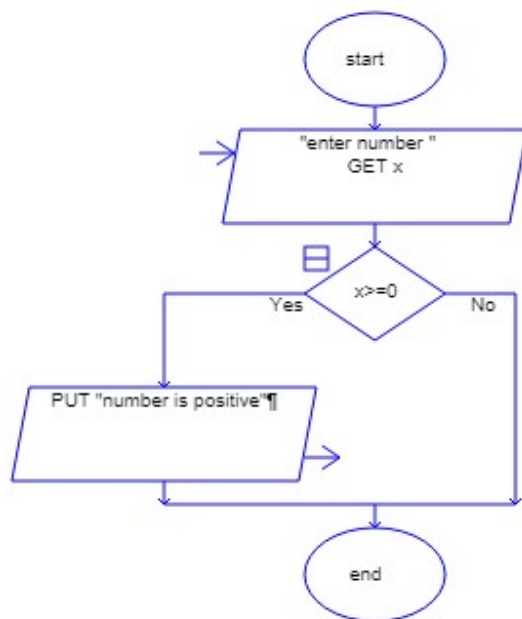


ASSIGNMENT NO: 01

1. Check Positive Number

- Task: Create a flowchart to check whether a number is positive.



- Next Step: Write a Java program that checks if a predefined number is positive using an if-else statement and prints the appropriate message.

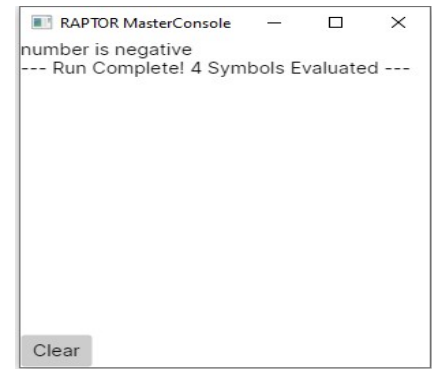
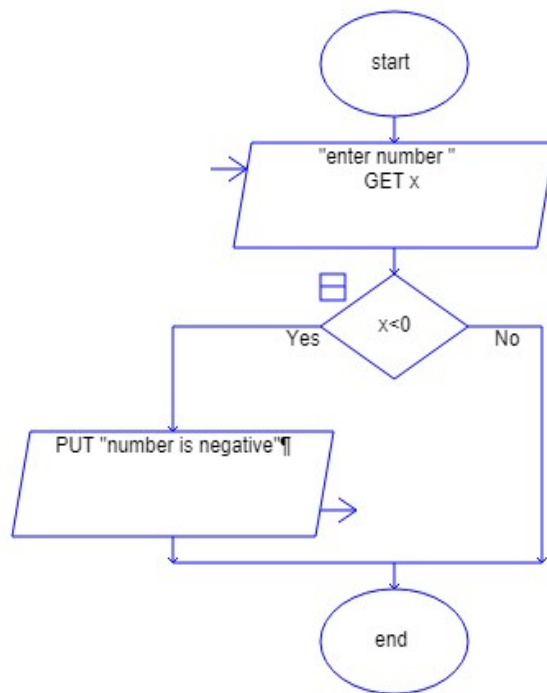
```
class sample
{
    public static void main(String args[]) {
        int x=8;
        if(x>=0)
            System.out.println("x is positive");
    }
}
```

Output:

x is positive

2. Check Negative Number:

- Task: Create a flowchart to check whether a number is negative.



- Next Step: Write a Java program that checks if a predefined number is negative using an if-else statement and displays the result

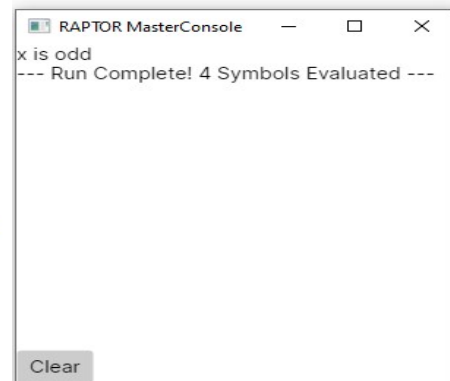
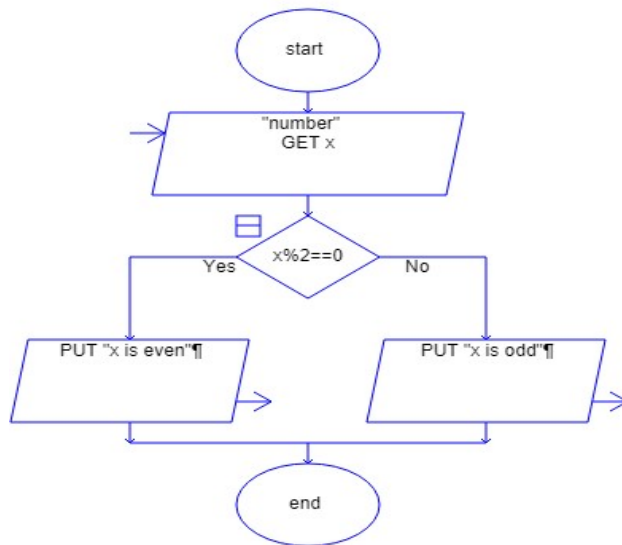
```
class negative
{
    public static void main(String args[]) {
        int x=-7;
        if(x<0)
            System.out.println("x is negative");
        }
    }
```

Output:

x is negative

3. Check Odd or Even Number:

- Task: Create a flowchart to determine whether a number is odd or even.



- Next Step: Write a Java program that checks if a predefined number is odd or even. Use an if-else statement and the modulus operator (%) to determine whether the number is divisible by 2 or not.

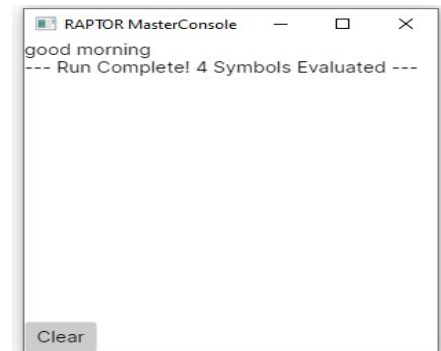
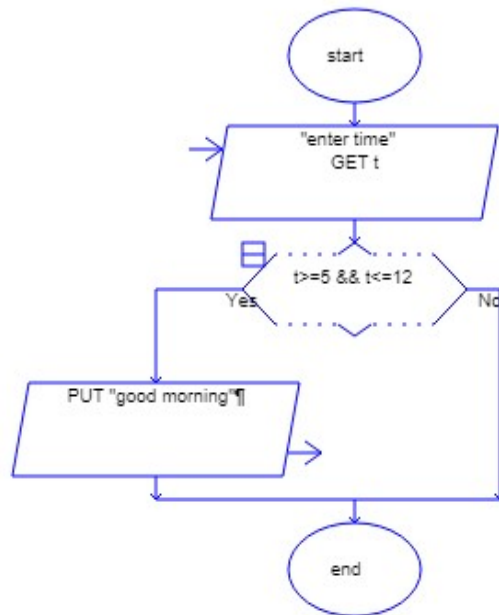
```
class evenorodd
{
    public static void main(String args[]) {
        int x=5;
        if(x%2==0){
            System.out.println("x is even");
        }
        else{
            System.out.println("x is odd");
        }
    }
}
```

Output:

x is odd

4. Display Good Morning Message Based on Time:

- Task: Create a flowchart to display a "Good Morning" message based on a given time.



- Next Step: Write a Java program that displays a "Good Morning" message if the predefined time is between 5 AM and 12 PM. Use an if statement to implement the logic.

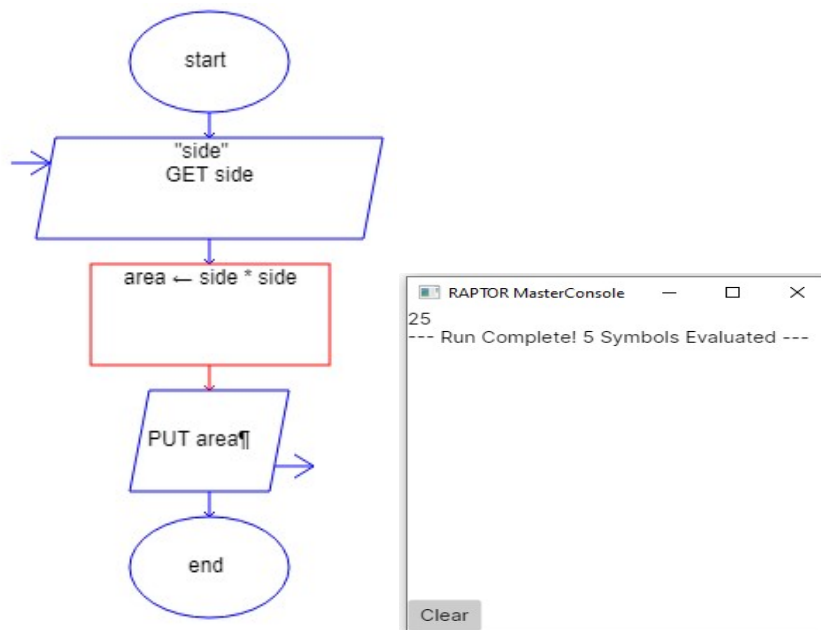
```
class goodmorning
{
    static void main(String args[])
    {
        //24 hour system
        int time =8;
        if(time>=5&&time<=12)
        {
            System.out.println("good morning");
        }
    }
}
```

Output:

good morning

5. Print Area of a Square:

- Task: Create a flowchart to calculate and print the area of a square.



- Next Step: Write a Java program that calculates the area of a square using the formula $\text{area} = \text{side} * \text{side}$. Use a predefined side length.

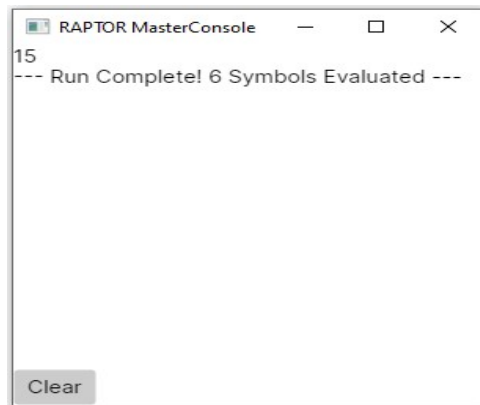
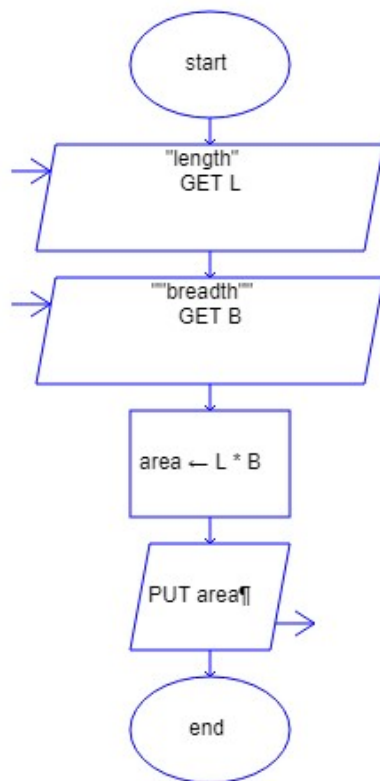
```
class areaofsquare{
    static void main(String args[]){
        int side=5;
        int area=side*side;
        System.out.println("area of square is "+area);
    }
}
```

Output:

area of square is 25

6. Print Area of a Rectangle:

- Task: Create a flowchart to calculate and print the area of a rectangle.



- Next Step: Write a Java program that calculates the area of a rectangle using the formula $area = length * width$. Use predefined values for length and width.

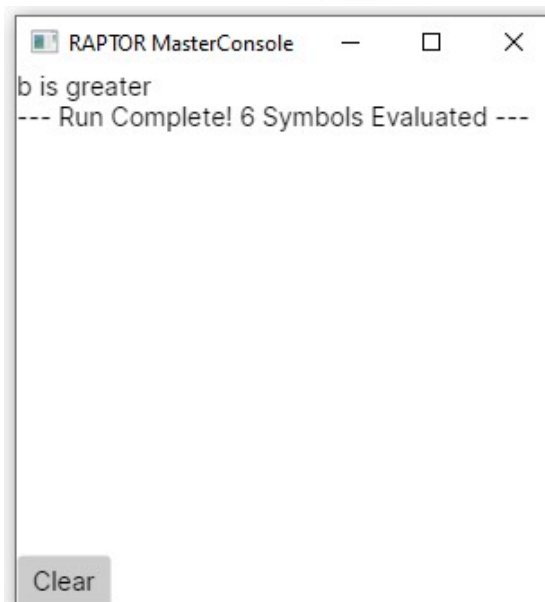
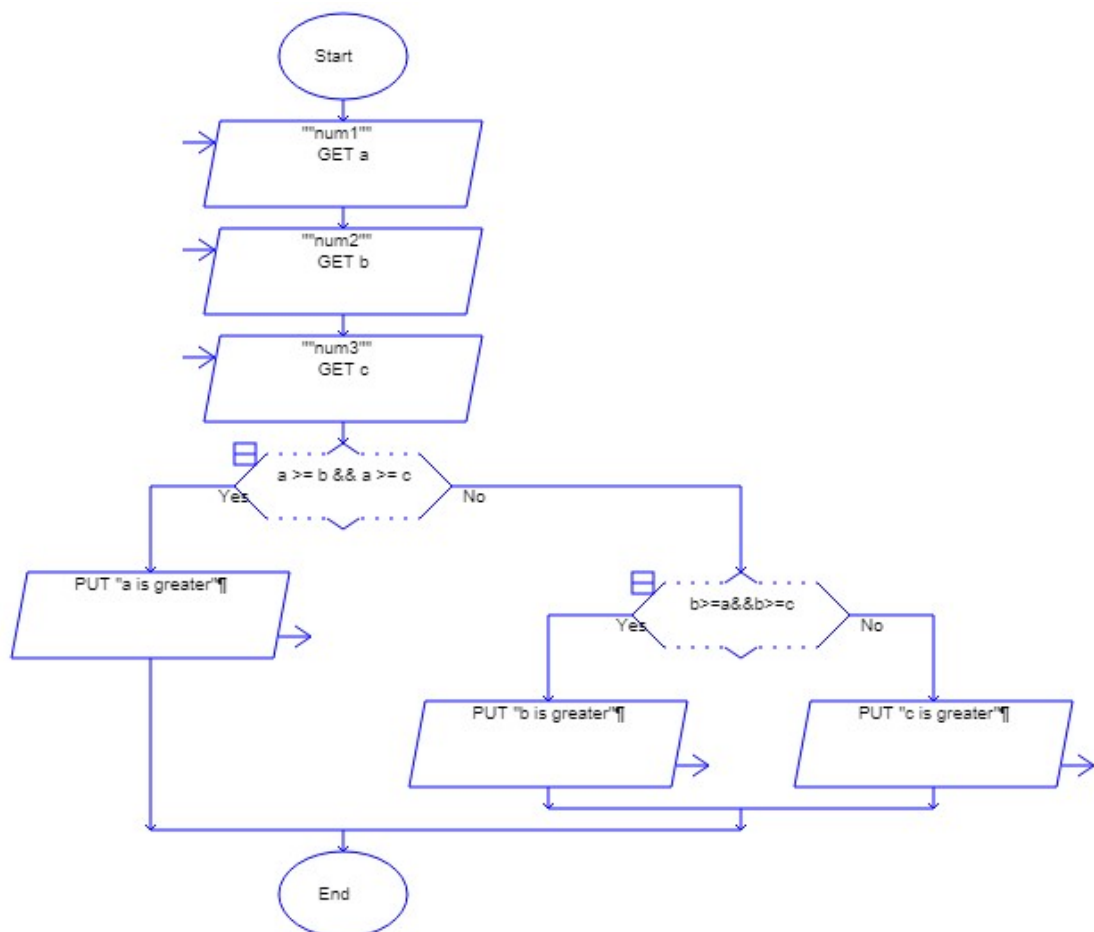
```
class areaofrectangle{
    static void main (String args[]){
        int l=5;
        int b=3;
        int area=l*b;
        System.out.println("area of rectangle is "+area);
    }
}
```

Output:

area of rectangle is 15

7. Find the Largest of Three Numbers:

- Task: Create a flowchart to find the largest of three numbers.



- Next Step: Write a Java program that finds and prints the largest of three predefined numbers using if-else statements.

```
class greaterofthree
{
    static void main(String args[]){
        int a=3;
        int b=8;
        int c=5;
        if(a>=b&&a>=c){
            System.out.println("a is greater "+a);
        }else{
            if(b>=a&&b>=c){
                System.out.println("b is greater");
            }else{
                System.out.println("c is greater");
            }
        }
    }
}
```

Output:

b is greater